

PRIYANSHU RANJAN

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 CodeChef

Education

Birla Institute of Technology, Mesra <i>B.Tech in Computer Science & Engineering (AI & ML)</i>	2023 – 2027 <i>CGPA: 8.12/10</i>
Surendranath Centenary School, Ranchi <i>Senior Secondary (CBSE) – 12th Grade</i>	2023 <i>Percentage: 89%</i>
Surendranath Centenary School, Ranchi <i>Secondary (CBSE) – 10th Grade</i>	2021 <i>Percentage: 96%</i>

Experience

Research Intern <i>BIT Mesra, Ranchi</i>	May 2026 – present
- Built a CNN-Transformer model for pre-apnea ECG classification on PhysioNet Apnea-ECG dataset, achieving 87%+ accuracy and 92%+ sensitivity via 5-fold cross-validation. - Engineered sliding-window segmentation pipeline with Z-score normalization and class balancing to predict apnea events 45 seconds before onset from single-lead ECG signals. - Optimized training with mixed precision and early stopping in PyTorch, cutting training time by ~2x on GPU.	

Projects

CloudIT – Cloud-Based File Sharing Application:	Source Code
- Built a scalable cloud-based file-sharing platform enabling secure uploads, link-based sharing, and efficient file management, designed for high availability using JWT authentication and Cloudinary integration. - Optimized backend APIs with Node.js and Express.js, achieving around faster local file transfer speeds through stream-based uploads and middleware optimization, with data protection via Bcrypt encryption. - Technologies: React.js, Node.js, Express.js, Tailwind CSS, JavaScript, JWT, Bcrypt, Cloudinary.	
SyncSpace – AI-Powered Real-Time Collaborative Whiteboard	Source Code
- Built a full-stack collaborative whiteboard with instant multi-user sync using Socket.io, React, and a Node.js backend. - Integrated Groq AI (LLaMA 3.3 70B) to generate architecture diagrams from prompts, converting JSON into Rough.js canvas elements in real time. - Implemented JWT authentication and socket-level authorization, ensuring only permitted users can access or edit shared boards. - Technologies: React.js, Node.js, Express.js, Socket.io, MongoDB, Rough.js, Groq API, JWT	

Skills

- Languages:** C++, Python, JavaScript, HTML
- Frameworks & Libraries:** Node.js, Express.js, ReactJS, Tailwind CSS
- Tools/Platforms:** Docker, Postman, GitHub, MongoDB
- Relevant Coursework:** Machine Learning, Data Structures, Computer Networks, Operating Systems, Object-Oriented Programming, Database Management Systems

Competitions & Achievements

- Solved **1000+ DSA problems** on **LeetCode, CodeChef** and **Codeforces**, achieving a LeetCode max rating of **1754**.
- Rated **2 Star** on CodeChef, achieving a global rank of **337** in **STARTERS 237**. [[CodeChef Profile](#)]
- Secured **Top 5 place** in the **IEEE Megaproject Hackathon**- a flagship hackathon of IEEE BIT Mesra..
- Secured a global rank of **3877** in a **LeetCode Weekly Contest** among **27,000+ participants**.
- Secured a global rank of **5947** in a **Codeforces Div. 2** contest among **28,000+ participants**.

Honours and Awards

- Secured **State Rank 09** in the prestigious **National Talent Search Examination (NTSE)**.
- Achieved **All India Rank 03** ,in Abacus Prodigy Competition , declared **State Champion** and **City Topper** among **10K+ students** nationwide.
- Secured **All India Rank under 50** in a **National-level Painting Contest** organized by **HDFC**.

Position of Responsibility

- Executive Member-IEEE BIT Mesra:** Conducted a **DSA workshop** for **200+ students**, introducing **competitive programming** and **data structures fundamentals**.